

Problem Statement

Loan Approval Prediction Using Machine Learning and Data Analytics

Financial institutions receive a large number of loan applications every day. Evaluating each application manually is time-consuming and may lead to inconsistent decisions. The approval of a loan depends on several factors such as applicant income, loan amount, CIBIL score, employment status, education, and asset values.

The objective of this project is to analyze historical loan application data and develop a predictive model that can determine whether a loan application is likely to be approved or rejected. The project uses **Excel, SQL, Python, Power BI, and Tableau** to perform data cleaning, analysis, visualization, and machine learning-based prediction.

The system helps banks and financial institutions make faster, more accurate, and data-driven loan approval decisions while reducing risk and improving operational efficiency.

Objectives

1. Analyze loan application data.
2. Identify factors influencing loan approval.
3. Build a machine learning model to predict loan approval status.
4. Visualize key insights using Power BI and Tableau dashboards.
5. Support data-driven decision-making for financial institutions.

Expected Outcome

A loan approval prediction system that classifies applications as **Approved** or **Rejected** based on applicant financial and personal details, along with interactive dashboards for business analysis.